

Avinash K. Goyal

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EDUCATION

Northeastern University / American College of Thessaloniki

Sept 2024 - May 2028

- Candidate for Bachelor of Science - Mechanical Engineering
 - Relevant Classwork: **Mechanics of Materials**, Statics, & Dynamics (In Progress)
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SKILLS & CERTIFICATIONS

- Software: **SolidWorks**, Blender, AutoCAD, Arduino IDE, & **Microsoft Suite**
 - Electronics & Programming Languages: **ESP32 & Arduino Microcontrollers**, Soldering, Java, & C++
 - Machinery & Manufacturing: **3D Printer**, CNC Machine, Laser Cutter, Hand Tools, & Band Saw
 - Certifications: **SolidWorks Certified Associate**
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PROJECTS

Digital Puzzle Box | Personal Project

Sept - Oct 2025

- **Designed and built an interactive, multi-stage puzzle box using an ESP32, LCD, distance sensors, buttons, joystick input, and servo activation**
- Programmed sequential logic puzzles in **C++ (Arduino IDE)**, including distance-based timing, binary decoding, and joystick input patterns
- Implemented embedded state-machine logic to manage puzzle progression, input validation, and actuator control
- Fabricated a functional prototype in under a two-week build timeline using rapid, low-cost materials

Plant Drum Spinner Combat Robot | Combat Robotics

May - June 2025

- **Designed and built a compact, 1 lb combat robot featuring a drum spinner weapon system and low-profile chassis**
- Modeled and created **assembly** for chassis, armor, forks, and weapon geometry in **SolidWorks**
- Designed for inverted operation, enabling continued mobility after flips or rollovers
- Engineered sloped armor and low-clearance forks to improve engagement and defensive performance
- Integrated electronic systems within tight spatial constraints, including custom soldered circuitry

Animatronic Alligator Snapping Turtle | High School Engineering Class

Jan - June 2024

- **Designed and fabricated a full-scale external body shell for an animatronic alligator snapping turtle, balancing realism, durability, and integration constraints**
 - **3D modeled and printed** all body components in Blender, ensuring compatibility with internal systems
 - Led iterative material and surface experimentation to replicate organic skin texture, flexibility, and appearance using flexible TPU printing, fabric bonding, and epoxy-based treatments
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WORK EXPERIENCE

E.P. Heller Company Machining Intern

May - June 2024

- Monitored the processes of 1 - 3 CNC machines simultaneously and loaded drill bits upon completion
 - Packaged 250 - 1000 finished tools during daily CNC production runs
 - **Disassembled, repaired, and tested** multiple types of air-powered tools
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HONORS

- 2023 TransOptions Hydrogen Fuel Cell Competition: **Best Engineered Car (2023)**
- Ranney School Vex Robotics Competition Over Under Tournament: **Quarterfinals (2023)**